

UZAIR AKBAR

SUMMARY

Software Eng. with experience in Data Eng., Data Science, and DevOps—focused on model deployment, scalable ETL pipelines, and CI/CD. Pursuing an M.Sc. in Computer Science at Georgia Tech, specializing in Robust + Causal ML,[†] with strong interest in Data/ML Eng. and cloud based micro-services. Open to roles in SF, Atlanta, or remote.
Location: Atlanta, GA | **Work Auth.:** F-1 OPT (valid through Dec. '26, extendable to Dec. '28).

EDUCATION

M.Sc. Computer Science @ Georgia Institute of Technology **GPA:** 3.78 / 4.0 | **Dec. 2025**

- **Teaching Assistant:** Reinforcement Learning ([CS8803](#), [CS7642](#) Fall '24 — present).
- **Coursework:** Machine Learning, Systems for ML, Database Systems, Computer Networks, OS, Algorithms

M.Sc. Communications Eng. @ Technical University of Munich, Germany **GPA:** 3.65 / 4.0 | **June 2021**

B.Sc. Electrical Eng. @ National University of Sciences & Technology, Pakistan **GPA:** 3.59 / 4.0 | **June 2017**

SELECTED PUBLICATIONS

[†]**U. Akbar et al.** “An Analysis of Causal Effect Estimation using Outcome Invariant Data Augmentation.” *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. **(Spotlight; top ~3%)** [🔗](#) [📄](#) [🗣️](#)

A. Savkin, T. Lapotre, K. Strauss, **U. Akbar** & F. Tombari, “Adversarial Appearance Learning in Augmented Cityscapes for Pedestrian Recognition in Autonomous Driving.” *ICRA*, 2020. [🔗](#)

PROFESSIONAL EXPERIENCE

Software Engineer Intern @ Hewlett Packard Enterprise, USA **June 2024 — Aug. 2024**

- Integrated dynamic KVM machine provisioning (`k8s` + `KubeVirt`) w/ static ESXi based QA infra. to cut costs.
- Built CI/CD pipeline (`jenkins` + `rabbitMQ` + `redis`) to test, build & release VM templates for QA workloads.

Software Engineer @ OMMA, Germany **Nov. 2022 — June 2023**

- Designed end-to-end Salesforce ETL pipeline (`Airflow` + `PostgreSQL`), improving data throughput by 58% (by refactoring `python` → `postgres`) and sales lead identification by 63% (by reducing data redundancy & duplicates).
- Deployed `FastAPI` + `Docker` based CRM microservices on AWS, including `XGBoost`-based ticket classification and RegEx based error code identification.

Data Scientist @ Auto1 Group, Germany **June 2022 — Oct. 2022**

- Engineered `LightGBM` predictor for car refurbishment costs; €160 MAE decrease (~27.5% improvement) vs. prod.
- Architected data pipelines to streamline model retraining and deployment using S3 + Redshift.

ML Research Intern @ Max Planck Institute of Intelligent Systems, Germany **Apr. 2020 — Feb. 2021**

- Proposed a novel use-case of data-augmentation for causal inference – published as my Masters [thesis](#) at TUM.
- Developed a method to reduce hidden confounding bias w/ data-augmentation + instrumental variable regression.

Data Science Intern @ PwC, Germany **Dec. 2019 — Apr. 2020**

- Trained Siamese LSTM/CNN (`PyTorch`), doc2vec (`Gensim`) for internal compliance doc. search/annotation app.
- Deployed the models as micro-services on AWS EKS with Seldon-core (containerization+API endpoint) & `k8s`.

ML Research Intern @ BMW Group, Germany **July 2019 – Nov. 2019**

- Published novel GAN architecture for sim→real augmentation ([ICRA '20](#)), improving segmentation by 6.5% AP.

Data Science Intern @ Allianz, Germany **Oct. 2018 — June 2019**

- Built `sk-learn` compatible `python` package for fraud detection in corporate insurance claims via outlier-detection.
- Implemented features for interpretability + explainability of fraud scores; `lime` based feature importance, etc.

TECH. STACK

ML & Data Sci.: `Python` `PyTorch` `sk-learn` `TensorFlow` `LLaMA2` `LightGBM` `XGBoost` `Tableau` `C++`

DevOps & Cloud: `Git` `Bash` `Docker` `K8s` `Jenkins` `RabbitMQ` `JFrog` `AWS` (EC2, S3, Redshift, EKS)

Data Eng. & Back End: `MySQL` `PostgreSQL` `Redshift` `Redis` `Airflow` `Spark` `FastAPI`